

Understand and explain early layers so we can start supplementing >>

We define the prompt as a sequence of m tokens, $u=(t_1, t_2, \dots, t_m)$. In the early layers of the Transformer, these tokens are projected by the static pre-trained weights (W_K, W_V) into the Key (HK) and Value (HV) Hilbert spaces:

$$k_i = W_K t_i$$

$$v_i = W_V t_i$$

In standard attention, we keep these as discrete vectors. But to view the model as dynamically generating "new neurons," we treat them as continuous operators, applying **prompting dynamics and in-context learning states**, through a rigorous geometric thinking.

Early Layers and the belief of “sequential” tokening

In a Transformer, positional encodings are **not** injected token by token in a temporal sequence.

Mathematically, the input X (the matrix of pure word embeddings) and the positional encoding P (the matrix of sine/cosine frequencies) are added together entirely in parallel:

$$X_{in} = X + P$$

This operates on the entire set simultaneously. In the language of our previous operator discussion, P is part of the initial unitary transformation $U(u)$ that prepares the entire superposition state ρ at once. There is no sequential "injection."

Sequential Signaling - Escaping the symmetry and rebuilding it. SSC Ready

Without positional encoding, a Transformer is strictly a **set-processing operation**, permutation invariant. In the context of the Hilbert space H and the metrizable space $C(X)$, Positional Encoding acts akin to a **symmetry-breaking mechanism**, projecting continuous, order-compatible manifold through the sine and cosine functions of the optimization phases, establishing geometric frequency on how concepts relate over different scales of distance, defining the baseline distance metric upon which the Attention Operator A acts.

Generating Transient Weights ($\Delta W(u)[i]$) - Caching the cache

Before analyzing this synthesis, it is necessary to establish the mathematical foundation....

Insert fundamentals for hilbert, transience details on HOPE and all

We rely on normalized interactions that try to preserve isometries, and induce a Cauchy-like convergence.

Noisy Quadratic Limit says that near the optimum, the loss surface is governed by a Taylor expansion (a polynomial), so we already in a quadratic of sorts, symmetric topology.

Causal Masking on Extended Hilbert Manifolds

At inference time T_x , a new query token q_x enters the system. - relates to the Taylor expansion quote - In a standard unrolled linear attention framework (abstracting away the softmax nonlinearity to isolate the topological geometry), the output o_x is computed by applying a transient operator directly to the query:

$$o_x = \Delta W(u) q_x$$

Because of **causal masking**, the prompt tokens 1 through m are fixed. The operator $\Delta W(u)$ has reached **Cauchy convergence**. It is no longer updating its foundational topology, emerging as a **Transient Weight Operator**.

Insert / improve quote directly from the *Hope* paper's definition of neurons as rank-1 operators in”
 $H_{in} \otimes H_{out}$, we assume **Transient Weight Operator**, $\Delta W(u)$ - Interpret it as projecting “new neurons”.

In this context space, the composition of continuous functions $(p(f_1 \dots f_n))$ intersects the continuous space $C(X)$ finitely, therefore, it is fair to assume that the "trajectory" of a prompt isn't chaotic or purely sequential, It follows a predictable, smooth topology, much like the quadratic bowl of the optimization landscape.

Instead of calculating attention sequentially, we define $\Delta W(u)$ as the sum of the outer products (the rank-1 operators) of the prompt's Keys and Values:

$$\Delta W(u) = \sum_{i=1}^m v_i v_i^T k_i^T = V_p T K_p$$

Where:

- $v_i v_i^T k_i^T$ is a single rank-1 operator (a "transient neuron").

- $\Delta W(u) \in HV^{\otimes} HK$ is the aggregated dynamic weight matrix instantiated entirely by the prompt u .

Therefore, the early layers are able to write a temporary, prompt-specific weight matrix $\Delta W(u)$ directly into the context space. - L1 deepseek comparison.

In other words, By time T_x , the attention mechanism does not need to traverse the raw sequence $t_1 \dots t_m$ to find its relations. The causal masking emerged, and the total effective weight of the layer has been temporarily shifted to:

$$W_{eff(l)} = W_{base(l)} + \Delta W(u)_{(l)}$$

Which generates induction enough to eliminate backward or forward extra passes, given that the metric tensor A uses $\Delta W(u)$ as a **prebuilt Hilbert manifold**. (The transient weights have already deformed the distance metric $d(x,y)$, meaning the memory (d_m) and task states (d_i) required to solve the prompt are now geometrically adjacent to the execution state (d_E)) -

— explain how this proves as an stablized matrix and emmegers through Causal Masking